SUBHASIS DATTA

Chief Data Scientist | Chief AI/Machine Learning Officer| Technical Project Manager | Gen AI & Agentic AI SME| Thought Leader |

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WORK STATUS: US CITIZEN; HELD DOD, DHS, MULTIPLE FEDERAL FULL BACKGROUND CLEARANCES

EXPERIENCE HIGHLIGHTS AND KEY ACCOMPLISHMENTS

- 25+ Years' Individual Contributor and Team Lead experience as Data Scientist, AI/ML SME and Lead, PM, Expertise in successful design, modeling, validation, deployment, performance analytics of over five dozen AI/ML, Big Data Advanced Analytics programs
- Led team of 24 AI & Data Engineers/Scientists on USCIS Signature Backlog Reduction AI Automated POC closing 500K+ cases, saving 200K+ hours, $5M in 6 months; Closed Millions of cases in 2 years, saving $10's Millions, 10X speedup, 50% process speedup
- Principal Innovation Scientist and R&D Lead in seven AI/ML R&D Innovation programs leading to development of new AI analytical algorithms, testing ideas and proof-of-concept, and product development leading to new services, products, program offerings
- Built, led, mentored Data Science, multiple AI/ML Centers of Excellence leading over 20 Data Scientists, Engineers with $5-10M operational budgets for successful delivery on $250M AI/ML programs, infrastructure collaborating with cross-functional teams
- Transformed as AI Lead, largest Federal AI/ML driven Fraud Detection System at IRS increasing revenue protection by 2,500% from $1B to $25B+ in 4 years. Built AI/ML Fraud system handling billions of transactions, scoring 200M returns at 90%+ accuracy
- Successfully Architected 1,000X Performance & Scalability Modeling & Analytics of Cloud based Infrastructure and Applications, Managed hundreds of millions in Cloud Infrastructure Budget, Operations, Schedule & Performance Metrics, Fraud Analytics
- Saved $10'sM Infrastructure costs, improved ITSM Ops by 10% for largest Federal IT Infrastructure Outsourcing contract at $1.5B. Led Enterprise Infrastructure teams: Data Center, Network, Server, Desktop Ops, Cybersecurity, Incident Management Operations
- Designed and scaled AI pattern analysis algorithms for largest Insider Threat Unauthorized Access Detection program built in 9 months mining Billions of audit trail transactions for abusive patterns. Program detected $100'sM in abuse, misdemeanor cases.
- Designed, Built, Deployed, as AI/ML Lead, $70M Air Force (AF) Predictive Readiness System employing AI, Fuzzy Logic to predict real time AF Readiness of Forces analyzing billions of metrics and geospatial data utilizing Modeling & Simulation, Visualization.
- Transformed, Improved infrastructure processing performance of 25M taxpayers by 25% for IRS $2B 'System of Systems' IT Operations. As Data Scientist, Chief Architect, designed Enterprise Architecture, built Simulation Model of IRS Tax Filing Systems processing $1.7+T/year for 200M taxpayers. Program improved Critical Success Factors, Performance Metrics by over 25%.
- Won $1B+ Funded contracts, $25B+ IDIQ contracts as Primary Technical Author. As AI/ML SME led C-Level Executives for 20 years

EDUCATION

Professional Education & Micro Masters, MIT, Probability & Data Science MicroMasters, MIT Agentic AI Intensive Cert. (2025-27)
Professional Education, Stanford University & DeepLearning.ai, Machine Learning, AI Agents, LangGraph, LangChain LLM AppDev, LLMOps, RAG, Agentic RAG, Llamaindex, AI Agentic Design Patterns, AutoGen, MCP, Functions, Tools, Agents, Anthropic (2025-27)
MS Systems Engineering, John Hopkins University (Expected 2027)
PhD (ABD Status) Management Science and Information Systems, University of Maryland
MS Management Science and Information Systems, University of Maryland
MBA Operations Research and MIS, Southern Illinois University
BS Chemical Engineering, Jadavpur University
Certifications Completed/On-going: ITIL, KMPro, AIOps, DevSecOps, Certified Generative AI Expert™, CAIE Certified Artificial Intelligence (AI and ML) Expert®, Forward Deployed Engineer, Certified Agentic AI Developer™, Claude Code/Work, ChatGPT Codex, GenAI LLMs and Fine Tuning, Agentic AI, Multi-Agent Orchestration, MCP Servers, LangChain, LangGraph, AI Engineering, Azure AI Fundamentals, AWS Cloud Practitioner, Amazon Bedrock, Amazon Q, Amazon Sagemaker, Responsible AI, Guardrails (2025-27)

CAREER SUMMARY

AI/Machine/Deep Learning	**Generative AI and Agentic AI**	**Project/Program Lifecycle Management**
C-Level Relationship Management	**Cross-Functional Team Leadership**	**Product Management**
IT Infrastructure and Performance Management	**Innovation Management**	**Enterprise Architecture & Design**
Solution Engineering with LLMOps, DevSecOps	**Cloud Computing & Analytics**	**Predictive & Data Modeling/Analytics**

Concept Plus, Technical Project Manager, US Air Force Programs 2025 - Present

As Technical Project Manager, lead a team of engineers to build the next generation Predictive Maintenance models and systems for the USAF Comprehensive Engine Management System (CEMS), Reliability Centered Maintenance (RCM), leveraging Predictive Analytics, Reliability Statistics and Failure Analysis, AI/ML and Optimization techniques, to enable optimal decision making for the USAF leveraging flying performance, engine components and parts maintenance data, factoring in scope, cost, schedule, supply chain, time in depot, deployment readiness, and other factors to prepare the USAF aircrafts for the best mission readiness posture.

Digital Management Inc (DMI), acquired GreenZone, and The Ambit Group, **a DMI Company**, Chief Data Scientist and Machine Learning Officer, AI/ML SME, Program Manager, Federal AI/ML Programs, Chief Solution Architect, AI/ML R&D Lead 2020 – 2025

As Program Manager, Chief Data Scientist and AI/ML Lead, led the SEC Chief AI Officer for enterprise-wide AI/ML Program and Lifecycle Management, Enterprise Architecture for AI/ML Programs in the Cloud and On-Premises. As Chief Data Scientist, AI/ML and Advanced Analytics Lead for USCIS CDO led a team of 24 Data Scientists, AI/ML and Data Engineers to address a range of AI/ML - based USCIS Case Management, Network, Graph Analytics, NLP programs, led Intelligent Automation using Robotics Process Automation, Risk and Fraud Advanced Analytics problems. As SEC Technical Lead providing Technical Program Management support to a range of Enforcement and Analytics programs across the enterprise on a host of platforms including AWS Cloud, On-Premises.

- Built a new AI/ML Program leading the SEC Chief AI Officer and the Strategy and Innovation Office to architect an Enterprise level AI/ML Program across all of SEC. Facilitated SEC Chief AI Officer to lead all AI/ML Programs, responsible for approving, guiding and managing all AI/ML projects, ensure they adhere to OMB AI Guidance, White House Executive Orders, adhere to Responsible AI and AI Risk Management principles including use of AI Guardrails, MLOps, Performance Testing & Monitoring.
- Designed the AI Governance, AI/ML standard processes including Lifecycle management of AI/ML projects, AI/ML Infrastructure in the Cloud, and lead all the AI/ML projects under a unified platform with approved enterprise AI/ML software, tools, processes, and skilled workforce to enable the SEC mission of financial oversight of all public corporations, investments, funds.
- Led team of 24 AI & Data Engineers/Scientists on USCIS Signature Backlog Reduction AI Automated POC closing 500K+ cases, saving 200K+ hours, $5M in 6 months; Closed Millions of cases in 2 years, saving $10's Millions, 10X speedup, 50% process speedup
- As Chief Data Scientist and AI/ML Expert/Thought Leader responsible for leading AI/ML Predictive Modeling, AI/ML Heuristics driven Streamline Case Processing Automation for USCIS Chief Data Officer (CDO) case closures for millions of backlog cases.
- Wrote AI/ML algorithms in Python, Rules in PySpark and Databricks using LLMOps and MLOps to build AI/ML, Advanced Analytics models for CDO. As Advanced Analytics PM for CDO, led a team of over 24 Data Scientists, AI/ML and Data Engineers, Analysts to address a range of Rules-driven Streamlining of Adjudicative processes, Statistical & Advanced Analytics, Network, Graph Analytics, NLP, Intelligent Automation using Robotics Process Automation, Risk and Fraud Advanced Analytics problems.
- As Technical Lead at SEC, provided Technical Program Management support through IT and Engineering lifecycle to a range of Enforcement and Analytics programs across the enterprise on a host of platforms including AWS Cloud, On-Premises, towards technical solutioning, systems architecture and engineering, Agile program and project management.
- Conducted Predictive Modeling using Time Series ARIMA modeling in Python for building MVP models for predicting workload at USCIS; facilitated scaling models several orders of magnitude in Databricks environment with Apache Spark and Netezza

Received Special Recognition Award from Chief Data Officer for Innovation, AI/ML Streamline Case Processing Excellence

Simple Technology Solutions LLC, Chief Data Scientist, Chief Artificial Intelligence and Machine Learning Officer 2019 – 2020

As Chief Data Scientist and Principal for AI/ML responsible for leading STS Enterprise wide practice in Data Science, Machine Learning, and AI across multiple Agencies including DHS leading team of Data Engineers and Data Scientists, Cloud Architects and Engineers in design, modeling, evaluation, testing and deploying scalable AI/ML, Natural Language Processing (NLP) models

- Design and build enterprise-wide Data Modeling for Data Lake, Data Science, AI/ML, NLP, Text Analytics models using Big Data Analytics platforms and tools such as Spark Ecosystem using PySpark, Python Libraries, IBM SPSS Modeler, Netezza Enterprise Data Mart/Lake/Warehouse for risk-based scoring for threat analytics, target identification, performance engineering for TSA

AI/ML, DATA SCIENCE, SOCIAL GOOD CITIZEN DATA SCIENTIST (2013 – PRESENT):

- Industry Lead, ACT-IAC AI Working Group comprising Chief AI Officers, AI/ML COE Leads from all Federal Agencies, Industry
- Member, NIST Public Working Group on Defining Big Data, Analytics and IoT Standards
- TPM and Social Good Data Scientist deploying Machine Learning, AI, Text Analytics/Mining, UN Standby Task Force
- Senior Data Scientist, Program Manager for Data Analytics, ML/AI in social good programs (DataKind, Data Without Borders)
- Senior Social Good Data Scientist, TPM for humanitarian, underprivileged mission support, Statistics Without Borders

Analytica LLC, Chief Data Scientist, Director of Machine Learning and Artificial Intelligence 2018 – 2019
Chief Data Scientist, Director of AI/ML responsible for identifying financial fraud, maleficence in working with Federal Financial Oversight Enforcement agency. Built end-to-end prototype for DHS Enforcement and Benefits fraud, Person-to-Person matching. Advice HHS/CMS on Program Safeguard Fraud and Abuse, SME to IRS in Cloud based Advanced Analytics, AI/ML applications.

- Designed End-to-End Architecture and Prototype for DHS Person-to-Person Matching, Enforcement and Benefits fraud; employing Cloud based Enterprise data ingest, SAS based Analytics, Big Data Infrastructure, Tableau and Visual Analytics to identify abuse
- Built Unsupervised Learning AI/ML models for Outlier detection and Insider Trading algorithms for financial fraud/abuse detection

SES Corporation, Technical Program Manager and Chief Data Scientist 2017 – 2018
Technical Program Manager and Chief Data Scientist for the largest Federal Infrastructure development, Integration and Analytics Program for the 2020 Decennial Census, US Census Bureau. Responsible for hundreds of Infrastructure Engineers (Cloud, On-Prem, Networking), DevOps and Security Engineers building thousands of Servers for capturing, managing and analyzing the largest data capture, management and dissemination of 2020 Census. Managed Program Schedule, Performance, Budget and Analytics, Metrics.

- Advised, designed Cloud based Enterprise Fraud Analytics with PostgreSQL, SAS, Python/R Analytics on Enterprise Data Lake
- Led Infrastructure, DevOps, Security Engineering team with deploying, integrating thousands of cloud servers, program schedule, performance management, metrics analytics on hundreds of millions in Infrastructure operations, hundreds of components ATOs

ICF International, Technical Program Manager, Chief Data Scientist, Technical Director, Innovation R&D Lead 2013 – 2017
ICF PM/Lead responsible for program planning, management, control, delivery and success of multiple federal agency Analytics and Infrastructure programs including HUD, USPS, DoS, others using ML/AI and Advanced Analysis with Big Data where applicable.

- Transformed as PM/Lead, one of the largest federal IT Infrastructure outsourcing programs valued at over $1.5B. Managed cross-functional teams with Prime contractors and Agency responsible for Data Center, Mainframes, mid-tier Servers, Network Operations, Desktop and Mobile computing assessing their Performance Metrics as guided by their SLAs. Monitored performance using real time metrics assessment of program, infrastructure, cost data captured throughout the infrastructure
- Saved over $100M through streamlining operations, gaining efficiencies in Data Center Mainframe, Servers transitioning from On Premises to Cloud infrastructure, WAN network, mobile and desktop device consolidation, migration of applications to the Cloud
- Conducted automated performance analytics of 100+ metrics for 12 SLAs and Survey Analytics leading to 10% improvement
- Automated and improved Incident Management of millions of incident records analytics for root cause analysis, infrastructure downtimes, bottlenecks using data and text mining to identify patterns to improve operations by 15%
- PI for Innovation Award on Open Data initiative for Federal agency financial analysis using Cloud based BI and Dashboard

Received several Innovation Excellence Awards, Program Performance and Business Development, Proposal Writing Awards.

The MITRE Corporation, Chief Data Scientist/ML Officer, R&D Lead, Systems Engineer, Technical Program Manager 2006 – 2013
Chief Data Scientist/Architect, Lead for a Portfolio of dozens of Analytical, Machine Learning (ML) and AI, Infrastructure and Systems Engineering, Enterprise Architecture and Design, BPR programs. Transformed in this TPM and Chief Data Scientist, Machine Learning

Scientist role dozens of Fraud Detection, Compliance Improvement, Modeling and Simulation and Analytics programs along with their associated ML/AI Infrastructure working in collaboration with diverse cross-functional teams and C-level Executives for 7 years.

- Transformed as PM, one of the world's largest 'Big Data' fraud detection system while increasing revenue protection by 2,500% from $1B Year 1 to $25B+ by Year 4. Facilitated Executive leadership, cross-functional teams in building a new AI/ML Electronic Fraud Detection System (EFDS), AI Mining Infrastructure billions of transactions, scoring 200M returns real-time at 90%+ accuracy.
- Transformed, Improved infrastructure processing performance of 25M taxpayers by 25% for IRS $2B 'System of Systems' IT Operations. As Chief Scientist/Architect, designed Enterprise Architecture, built Simulation Model of IRS Filing Season Systems that processes $1.72T per year for 200M taxpayers. Program improved Critical Success Factors, Performance Metrics by 25%.
- Improved $2B IRS IT Infrastructure Operations by reducing downtime by 10% through deep understanding of Top 10 issues among 10% Incidents causing 90% of downtime, outages. As TPM and Data Scientist performed Data Analytics and Text Mining on millions of Incident Management records to analyze Help Desk Incidents, Patterns, Root Causes, Solutions to reduce risk.
- Reduced risk in 50 mission critical systems (multi-billion budget) by 20% by designing a comprehensive risk and readiness assessment framework with multi-dimensional factors and criteria to evaluate operational readiness and address key risks
- Improved operational system performance of $ multi-billions by 25% by developing end-to-end IRS 'System of Systems' architectural diagrams, dynamic Simulation Model for C-level Executive what if analysis scenarios for critical decision making
- As PI and R&D Lead won six Innovation proposals worth $1.5M over 10 years to build new capabilities, services and products

Received MITRE's highest recognition 'Directors Awards' twice and MITRE Corporate Knowledge Management President's Award for excellence in program and IT management

Alion Science & Technology, Asst. VP, Director for Center for Data Science and Systems Engineering, Sr. Data Scientist 1996 – 2006
Director of Data Science and Technical Program Manager, AI/ML Center for Excellence leading over 20 Data Scientist, Engineers and $5-10M line budget leading to successful delivery of $200M ML/AI programs and infrastructure. Collaborated with cross-functional teams to successfully impact and deliver on dozens of Analytical, Fraud Detection, Systems Engineering, AI/ML programs worth millions of dollars. As Director, TPM, acquired four product companies, startups to integrate their products as part of portfolio of products in $ multi-million programs and products sold. Created Fraud/Compliance Revenue Maximization ML/Analytics algorithms.

- Designed analytical and pattern analysis algorithms for one of the largest Insider Threat Unauthorized Access Detection program mining billions of audit trail transactions for abusive patterns mandated by Congressional law. Built engineering team and scaled algorithms and infrastructure as TPM for delivery within 9 months. Program detected $100'sM in abuse, misdemeanor cases.
- Designed, Built, Deployed as Lead, $70M Air Force (AF) Predictive Readiness System employing AI, Fuzzy Logic to predict real time AF Readiness of Forces analyzing billions of metrics and geospatial data utilizing Modeling & Simulation, Visualization
- As Chief Architect, Lead, designed DHS Incident Command and Director National Intelligence Information sharing Architecture
- Transformed Federal Crop Insurance Program by developing a new Fraud Detection program that detected $100'sM in new fraud protected and trained analysts and investigators in Data Mining and Analytical techniques. As Individual contributor and Lead of cross-functional teams designed over 25 analytical, ML, pattern recognition fraud detection algorithms to identify fraud.
- Won $1M+ R&D funds as PI/Data Scientist to Innovate on Analytical/ML algorithms for proof-of-concept tax revenue maximization product in collaboration with Oracle, Teradata to build Data Warehouse, 20 non-compliance detection algorithms.

Received 'Presidents Award' from IRS Commissioner and more than 12 Excellence awards from Alion Science and Technology for excellence in program, mission, project, and program performance

Additional positions:

- **University of Maryland**, **Chief Data Scientist, Director of Research and Institutional Studies, Information Systems**
- **Adjunct Faculty, Robert H. Smith School of Business – Taught Undergraduate and Graduate Computer Science and Business School courses**